Power BI: Adventure Works Sales & Returns Analysis

1. Executive Summary

Role: Lead Business Intelligence Analyst

Tools: Microsoft Power BI, Power Query, DAX

Dataset: Adventure Works (Global Manufacturing Company)

The **Adventure Works Sales Dashboard** is a comprehensive business intelligence solution designed to provide strategic insights into the performance of a global bicycle manufacturing company from 2020–2022. By integrating data from sales, customers, products, and returns, this project delivers a 360-degree view of business operations.

The final report empowers stakeholders to track key performance indicators (KPIs), analyze regional trends, and identify high-value customers to drive data-driven growth strategies.

2. Business Problem & Objectives

The management team at Adventure Works needed a way to track KPIs (sales, revenue, profit, returns) and compare regional performance. The raw data was scattered across multiple CSV files (Transactions, Returns, Customers, Products, Territories), making it difficult to spot trends or identify root causes for profit loss.

Key Objectives:

- Transform raw data into a professional-quality data model.
- Track core business KPIs (Revenue, Profit, Returns).
- Compare performance across regions, products, and customer segments.
- Identify high-value customers and trends in product returns.

3. The Process (Technical Implementation)

Step 1: Data Extraction & Transformation (ETL)

- **Import:** Connected to raw CSV source files (Sales, Returns, Products, Customers, Territories).

- **Transformation (Power Query):**
    - Cleaned and transformed data to ensure accuracy (removed duplicates, handled missing values).
    - Standardized data types and column headers.
    - **Date Table:** Created a dynamic rolling calendar table using Power Query M-code to enable time intelligence (Year, Quarter, Month, Week).
    - **Conditional Columns:** Added columns for customer categorization (e.g., Income Level, Parent Status) to enable demographic grouping.

Step 2: Data Modeling

- Designed a robust **Star Schema** data model optimized for performance.
- Established **One-to-Many relationships** between the Fact tables (Sales, Returns) and Dimension tables (Products, Customers, Calendar, Territories).
- Configured filter directions to ensure accurate propagation of context across the report.
- Hid foreign keys and technical columns to maintain a clean report view for end-users.

Step 3: Advanced DAX Calculations

Created calculated measures using DAX (Data Analysis Expressions) to move beyond simple aggregations:

- **Core Measures:** `Total Revenue`, `Total Cost`, `Total Orders`, `Total Profit`.
- **Time Intelligence:** Calculated `YTD Revenue` (Year-to-Date) and `Previous Month Revenue` to track growth.
- **Complex Ratios:** Created `Return Rate %` and `Profit Margin %` to assess efficiency, not just volume.
- **Dynamic Measures:** Utilized `ALL()` and `CALCULATE()` to create % of Total comparisons.

Step 4: Dashboard Design & Visualization

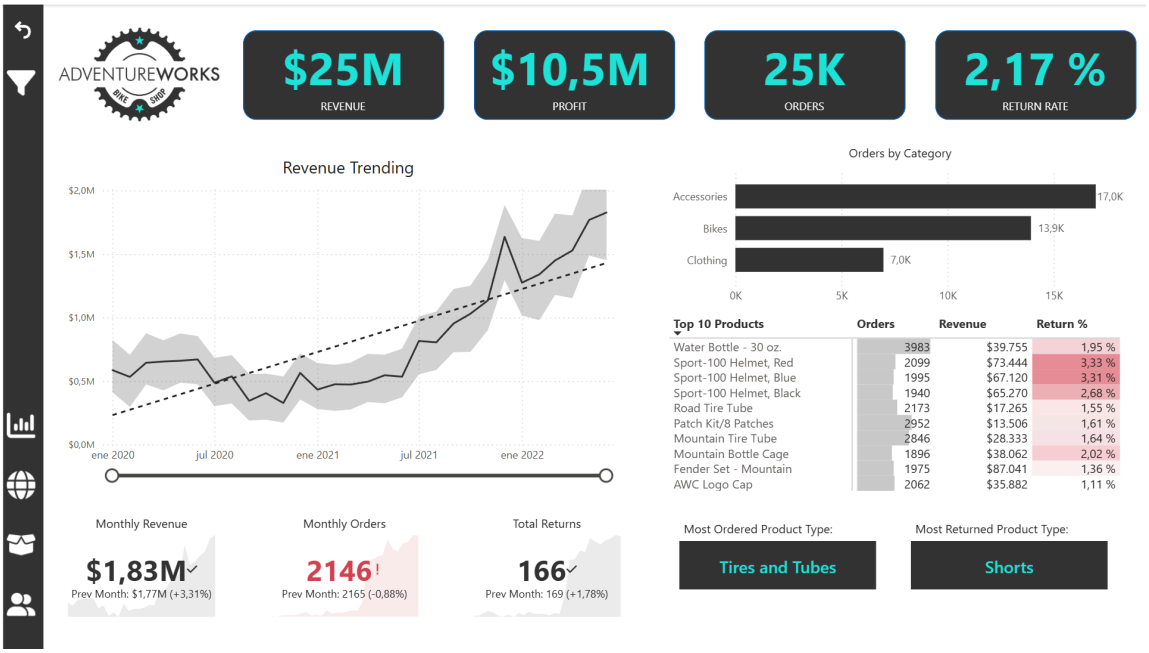
Designed an interactive report with 4 dedicated pages, using a cohesive color theme and intuitive navigation:

1. **Executive Dashboard:** High-level KPI cards, revenue trends by month, top 3 categories with more orders and top 10 products.
2. **Map & Territory Analysis:** Geospatial analysis showing sales distribution across North America, Europe, and the Pacific.
3. **Product Analysis:** This is a drillthrough page so users can navigate with pre-filtered on item selected. It contains detailed performance metrics comparing monthly orders, revenue, profit vs target, including a numeric parameter where you can see total profit based on the price adjustment %. Moreover, it includes field parameters to investigate main KPIs, such as, total orders, revenue, returns and return%.

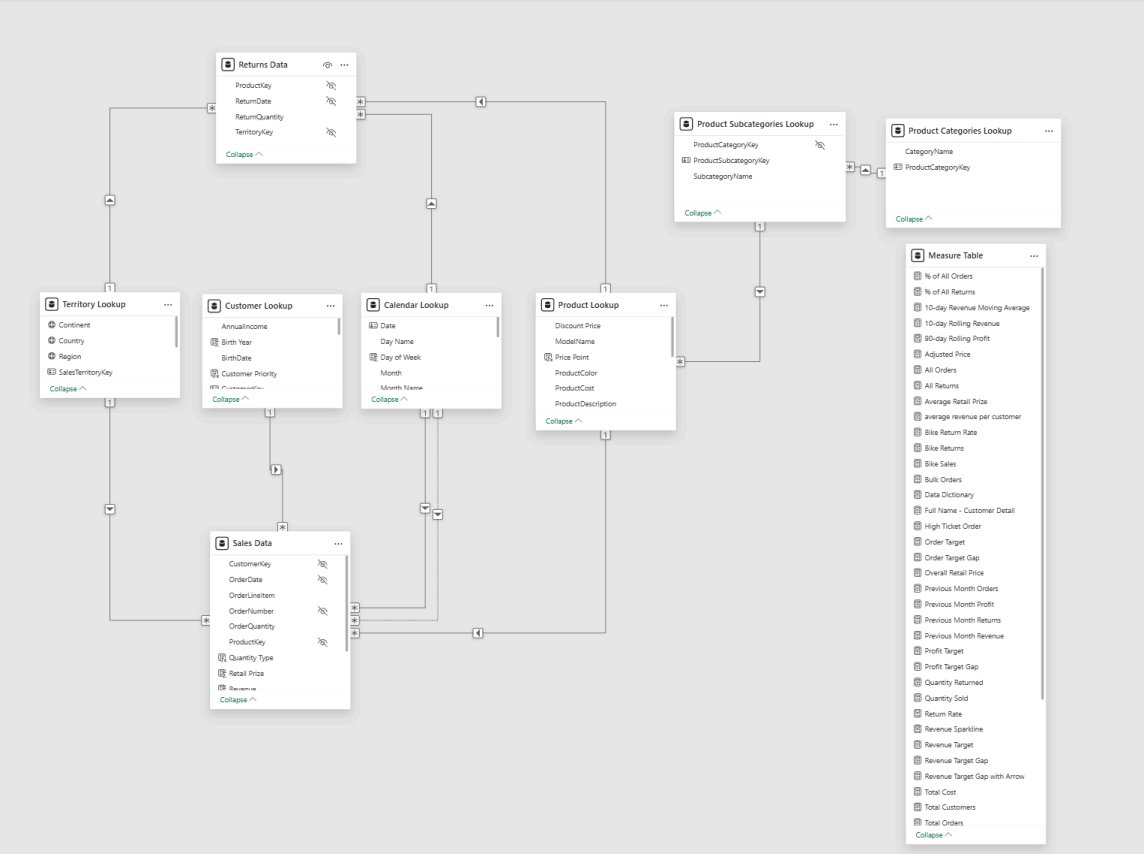
4. **Customer Analysis:** Segmentation of customers by income, occupation to identify the "ideal customer" profile. Moreover it includes trends on total customers and revenue per customer.

4. Visuals

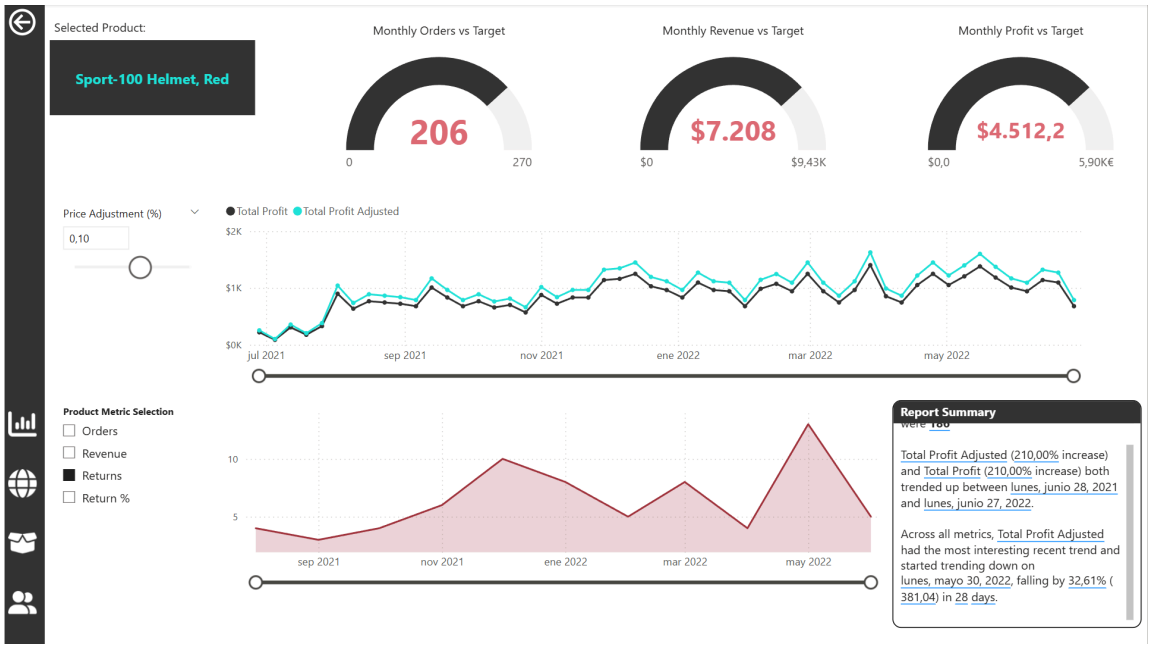
- *Screenshot 1: The Executive Dashboard showing the main KPIs.*



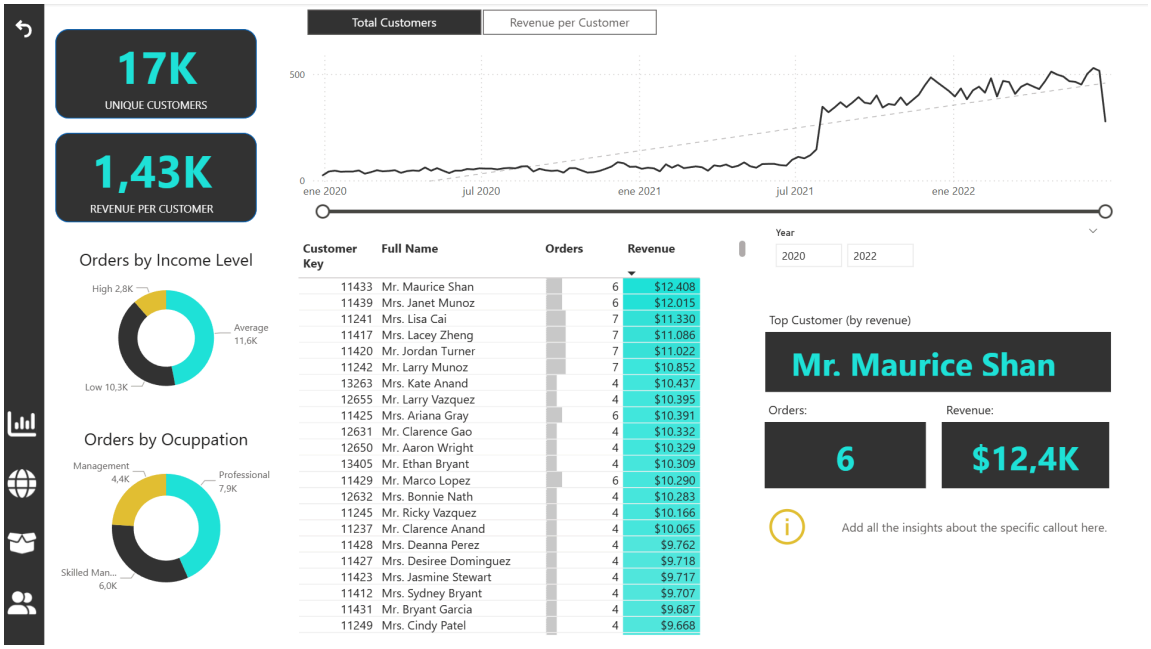
- *Screenshot 2: The Data Model (Star Schema) view.*



- *Screenshot 3: The Product Detail page showing numeric range parameters to view the total profit adjusted and KPI parameters that will automatically update the graph based on metric selected.*



- *Screenshot 4: The Customer Detail page*

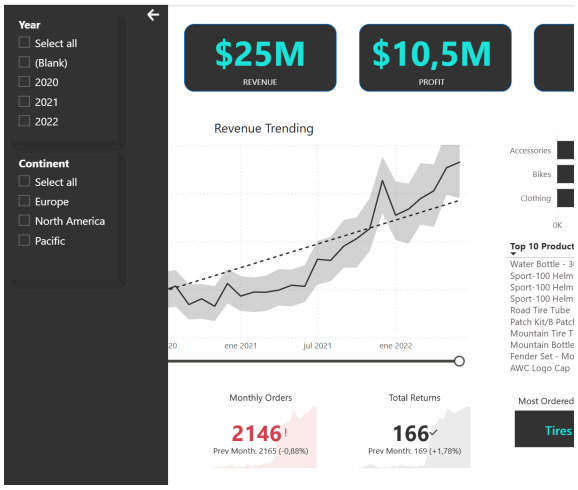


5. Advanced Dashboard Design & User Experience (UX)

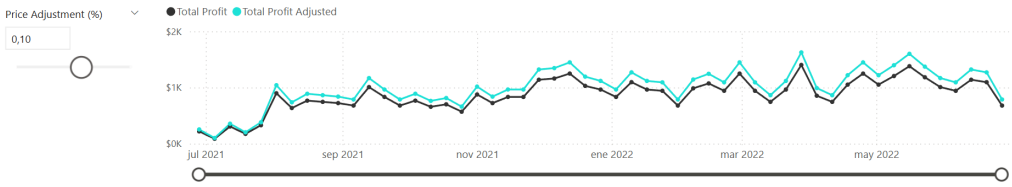
Beyond standard visualizations, I implemented advanced Power BI features to create an "app-like" navigation experience and deep analytical capabilities:

- **Custom Navigation & UI:**

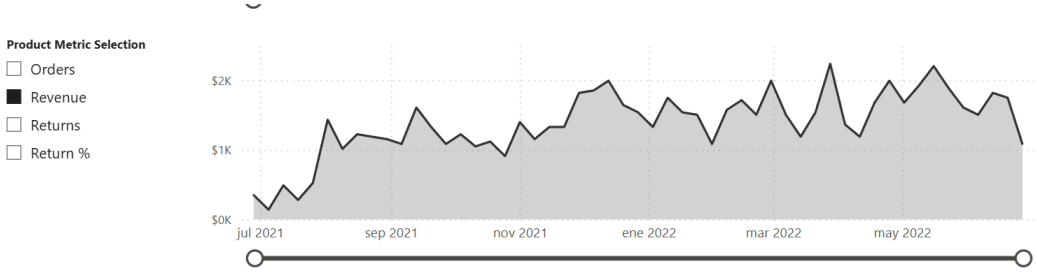
- Designed a custom **Page Navigation Bar** using buttons and bookmarks, allowing users to switch contexts (Executive, Map, Product) seamlessly without using default tabs.
- Implemented a **Collapsible Slicer Panel** using Bookmarks and the Selection Pane to add extra filters. This keeps the canvas clean by hiding filters when not in use, maximizing space for insights.



- **Interactive Analysis Tools:**
  - **Drill-Up & Drill-Down:** Enabled hierarchical exploration on date fields (Year > Quarter > Month) and geographic data (Country > State > City) for granular analysis.
  - **Drillthrough Filters:** Configured a dedicated "Product Detail" page. Users can right-click any product category on the Executive Dashboard to "drill through" and see specific metrics for that individual item.
  - **Custom Tooltips:** Designed report page tooltips that appear on hover. For example, hovering over a top category bar reveals a mini-chart of weekly orders and top KPIs, providing instant context without leaving the view.
- **Dynamic Parameters (What-If & Field Parameters):**
  - **Numeric Range Parameters:** Created a "Price Adjustment" slider (What-If Analysis) allowing stakeholders to simulate how a 10-20% price increase would impact Total Revenue and Profit.



  - **Field Parameters:** Added a dynamic slicer that allows users to toggle visual axes between "Orders", "Revenue", "Returns" and “Returns %” on the fly, reducing the need for multiple duplicate charts.



Step 6: Advanced Analytics & Governance

- **AI-Powered Insights:**
  - Applied **Anomaly Detection** to time-series line charts to automatically flag irregular spikes or dips in revenue, helping users spot outliers immediately.
- **Report Governance (Row-Level Security):**
  - Implemented **Row-Level Security (RLS)** using DAX roles.
  - Scenario: Configured a "Europe Manager" role that filters the report data automatically so that European stakeholders see only their region's performance upon logging in.
- **Visual Interactions:**
  - Managed **Edit Interactions** to prevent specific slicers from filtering unrelated visuals (e.g., ensuring the "Total Revenue" card remains static while the user filters the "Monthly Trend" chart), maintaining a stable reporting view.